# Lecture 4: ADMM Algorithm for Nonconvex Problems: Applications and Convergence Analysis


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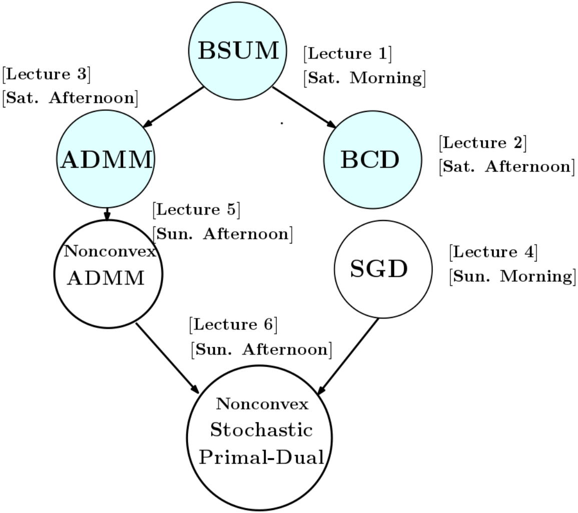
# Schedule (temporary)



# Main references (tutorials)

- M. Hong, Z.-Q. Luo, and M. Razaviyayn, "Convergence analysis of alternating direction method of multipliers for a family of nonconvex problems," SIAM Journal On Optimization, vol. 26, no. 1, pp. 337-364, 2016

# Agenda

- Review a few recent applications of ADMM on nonconvex problems
- A new analysis for nonconvex ADMM
  1. Basic proof steps
  2. Extensions
- Road map ahead

# Outline

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# The Basic Setup

- Consider the following linearly constrained (possibly nonsmooth or/and nonconvex) problem with $K$ blocks of variables $\{x_k\}_{k=1}^K$:

$$\begin{aligned} \min \quad & f(x) := \sum_{k=1}^{K} g_k(x_k) + \ell(x_1, \cdots, x_K) \\ \text{s.t.} \quad & \sum_{k=1}^{K} A_k x_k = q, \; x_k \in X_k, \; \forall\, k = 1, \cdots, K \end{aligned} \tag{1}$$

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- Here $A_k \in \mathbb{R}^{M \times N_k}$ and $q \in \mathbb{R}^M$; $X_k \in \mathbb{R}^{N_k}$ is a closed convex set; $\ell(\cdot)$ is a smooth (possibly nonconvex) function; each $g_k(\cdot)$ can be either a smooth function, or a convex nonsmooth function

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- Define $A := [A_1, \cdots, A_k]$

# The Basic Setup

- Note that this setup is more general than the one we saw in the previous lecture [Lecture 3]
  1. We have $K$ blocks of variables instead of two
  2. The smooth function $g$ couples all the block variables
  3. The smooth function can be nonconvex
- We cannot expect that the classical analysis of ADMM will work

# The Basic Setup

- The augmented Lagrangian for problem (1) is given by

$$L(x;y) = \sum_{k=1}^{K} g_k(x_k) + \ell(x_1, \cdots, x_K) + \langle y, q - Ax \rangle + \frac{\rho}{2}\|q - Ax\|^2,$$

where $\rho > 0$ is a constant representing the primal stepsize.

**Algorithm 0. ADMM for Problem** (1)

At each iteration $t+1$, update the primal variables:

$$x_k^{t+1} = \arg\min_{x_k \in X_k} L(x_1^{t+1}, \cdots, x_{k-1}^{t+1}, x_k, x_{k+1}^t, \cdots, x_K^t; y^t), \ \forall\ k. \quad (2)$$

Update the dual variable:

$$y^{t+1} = y^t + \rho(q - Ax^{t+1}). \quad (3)$$

# Overview of Applications

- ADMM has been widely (and wildly) used to solve nonconvex problems
    1. nonnegative matrix factorization [Sun-Fevotte 2014]
    2. phase retrieval [Wen-Yang-Liu-Marchesini 12]
    3. distributed matrix factorization [Ling-Yin-Wen 12]
    4. distributed clustering [Forero-Cano-Giannakis 11]
    5. polynomial optimization/tensor decomposition [Jiang-Ma-Zhang 13, Livavas-Sidiropoulos 14]
    6. matrix separation/completion [Xu-Yin-Wen-Zhang 11]
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- Even for discontinuous problems

- However, no formal convergence analysis is known

- Most analysis depends on uncheckable conditions such that $\|x^{t+1} - x^t\| \to 0$, $\|y^{t+1} - y^t\| \to 0$, etc.

# Application 1: NMF

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$$\min \frac{1}{2}\|VW - C\|_F^2, \text{ s.t. } V_{ij} \ge 0, \ W_{ij} \ge 0 \tag{4}$$

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- Introduce a new variable $Z$, we have the equivalent form

$$\min\ \frac{1}{2}\|Z - C\|_F^2,\ \text{s.t. } Z - VW = 0,\ V_{ij} \geq 0,\ W_{ij} \geq 0 \tag{5}$$

# Application 1: NMF (cont.)

- The ADMM iteration is given by

$$
\begin{aligned}
(Z^{t+1}, V^{t+1}) &= \arg\min_{Z,V\geq 0} \left(\|Z - C\|_F^2 + (\rho/2)\|Z - VW^t + U^t\|\right) \\
W^{t+1} &:= \arg\min_{W\geq 0} \|Z^{t+1} - V^{t+1}W + U^t\|_F^2 \\
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- "The second splits in the columns of W, so can be performed by solving parallel quadratic programs to find each column"
- Subproblems are NOT in closed form
- No convergence analysis!

# Application 2: NMF with Missing Values

- This formulation comes from [Xu-Yin-Wen-Zhang 11]
- Consider the following problem

$$\min_{X,Y} \|P_\Omega(XY - M)\|_F^2 \ \text{s.t.}\ X_{ij} \ge 0,\ Y_{ij} \ge 0 \tag{6}$$

- $\Omega$ indexes the known entries of $M$
- $P_\Omega$ is the projection to $\Omega$
- Introduce three new variables $Z, U, V$ (note $X$, $Y$ are coupled in the objective)

$$\begin{aligned}\min_{X,Y,Z,U,V} \ & \|XY - Z\|_F^2 \\ \text{s.t.}\ & U_{ij} \ge 0,\ V_{ij} \ge 0, \quad X = U,\ Y = V, \\ & P_\Omega(Z - M) = 0\end{aligned}$$

- Introduce multipliers $\Lambda$ (for $U$) and $\Pi$ (for $V$)

# Application 2: NMF with Missing Values (cont.)

- All the primal updates can then be written down in closed-form

$$
\begin{aligned}
X^{t+1} &= (Z^t(Y^t) + \alpha U^t - \Lambda^t)(Y^t(Y^t)^T + \alpha I)^{-1} \\
Y^{t+1} &= \left((X^{t+1})^T X^{t+1} + \beta I\right)^{-1}\left((X^{t+1})^T Z^t + \beta V^t - \Pi^t\right) \\
Z^{t+1} &= X^{t+1}Y^{t+1} + P_\Omega\left(M - X^{t+1}Y^{t+1}\right) \\
U^{t+1} &= P_+\left(X^{t+1} + \Omega^t/\alpha\right) \\
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- **Convergence analysis** If the successive differences of all the primal and dual variables go to zero (e.g., $X^{t+1} - X^t \to 0$ and so on), then any accumulation point of the iterates is a KKT point of the reformulated problem

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- This algorithm has been extended in [Sun-Fevotte 14] to general $\beta$-divergence criteria for NMF; Also extended to Tensor Decomposition [Liavas-Sidiropoulos 14]; Again closed-form updates,

# Application 3: Matrix Low Rank Factorization

- Consider the following formulation of low-rank+sparse matrix factorization

$$\min_{U,V} \|P_{\Omega}(UV - D)\|_1 \tag{7}$$

where $U \in \mathbb{R}^{m\times k}$ and $V \in \mathbb{R}^{k\times n}$ and $k$ is small

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- Write down the equivalent formulation

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- Again closed form solution; convergence based on assumption on diminishing distance between iterates

# Application 3: Matrix Low Rank Factorization (cont.)

- A related algorithm tries to compute a matrix low-rank factorization in a decentralized why [Zhang-Kowk 14]
- Consider the matrix low-rank factorization problem

$$\min_{X,Y} = \frac{1}{2}\|UV - D\|_F^2 = \sum_{m=1}^{M} \|UV(m) - D(m)\|_F^2 \tag{9}$$

  where $V(m)$ and $D(m)$ is the $m$th column of $V$ and $D$
- Each $V(m)$ and $D(m)$ is private, only available to a single agent $m$
- Equivalently

$$\min_{U,V,Z} = \sum_{m=1}^{M} \|Z_m V(m) - D(m)\|_F^2, \text{ s.t. } Z_m = U, \ \forall \ m \tag{10}$$

# Application 4: Finding Intersection of Two Sets

- This type of applications tries to find the intersection of two (possibly nonconvex sets) [Wen-Yang-Liu-Marchesini-12]
- Problem Statement

$$\text{find } x \text{ and } y, \quad \text{s.t. } x = y,\ x \in X,\ y \in Y. \tag{11}$$

- ADMM iteration is very simple

$$\begin{aligned} x^{t+1} &= P_X\left(y^t - \lambda^t\right) \\ y^{t+1} &= P_X\left(x^{t+1} + \lambda^t\right) \end{aligned}$$

# Outline

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# A Toy Example

- First consider the following toy nonconvex example

$$\min \quad \frac{1}{2}x^T Ax + bx, \quad \text{s.t.} \quad x \in [1, 2]$$

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- Consider the ADMM iteration with the update sequence $x \to z \to y$

# A Toy Example (cont.)

- Convergence?
- Randomly generate the data matrices $A$ and $b$
- Plot the following
  1. Primal feasibility gap $\|z - x\|$
  2. The optimality measure $\|x - \text{proj}\left[x - (Ax + b)\right]\|$
  3. The $x$-feasibility gap $\|x - \text{proj}(x)\|$
- The algorithm converges to a KKT point iff all three quantities go to zero

# A Toy Example (cont.)

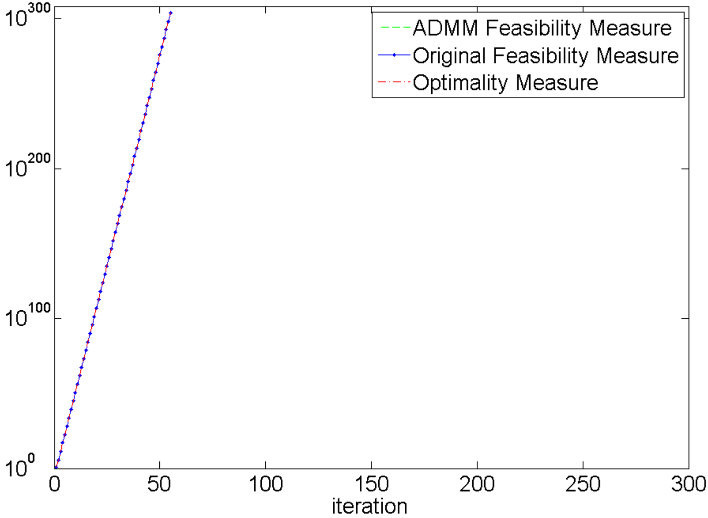



Figure: $n = 10$, $\rho = 20$

# A Toy Example (cont.)

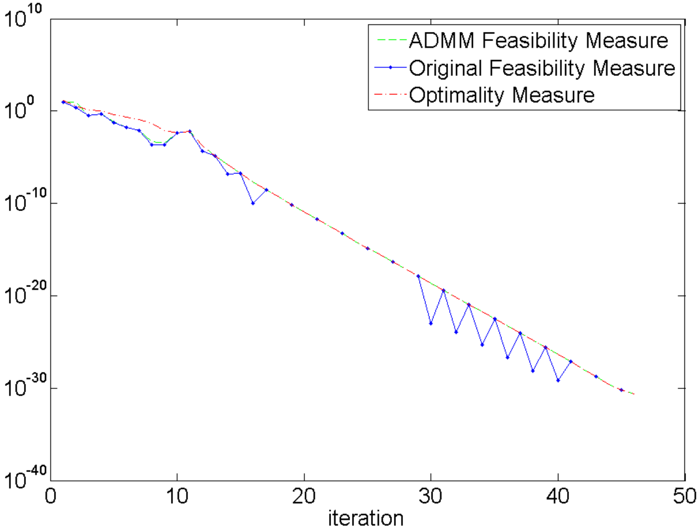



Figure: $n = 10$, $\rho = 200$

# A Toy Example (cont.)

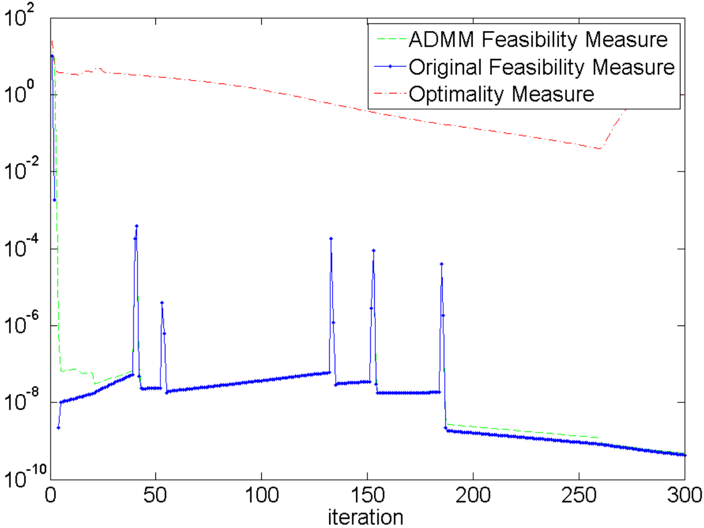



Figure: $n = 10$, $\rho = 2000$

# A Toy Example (cont.)

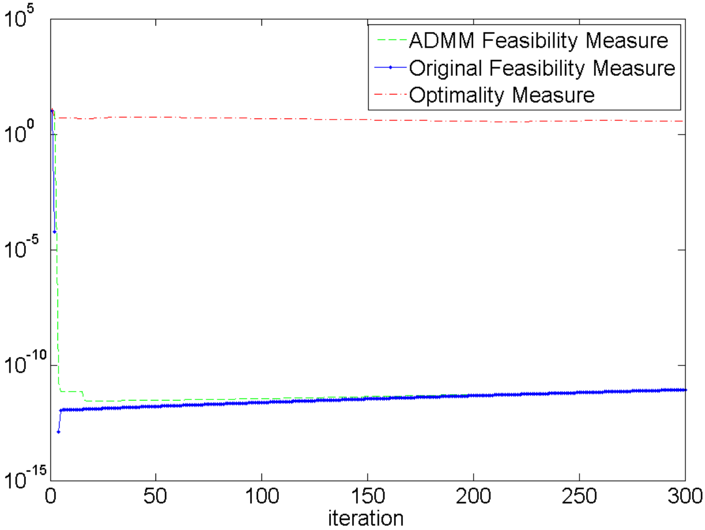



Figure: $n = 10$, $\rho = 20000$

# A Toy Example (cont.)

- The convergence is $\rho$-dependent
- When $\rho$ is small, the algorithm fails to converge
- When $\rho$ is large, maybe convergent
- Different from the convex cases (where $\rho$ does not affect convergence)

# Problem Setup: A Nonconvex Consensus Problem

- Consider a nonconvex global consensus problem

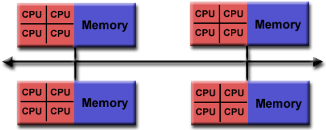
$$\min \quad f(x) := \sum_{k=1}^{K} g_k(x) + h(x) \text{ s.t. } \quad x \in X \tag{12}$$

- $g_k$: smooth, possibly nonconvex function
- $h(x)$: convex nonsmooth regularization term

# Problem Setup: A Nonconvex Consensus Problem

- Each $g_k$ needs to be handled by a single agent

$$\min \sum_{k=1}^{K} g_k(x_k) + h(x) \text{ s.t. } \quad x_k = x, \ \forall \ k = 1, \cdots, K, \quad x \in X. \tag{13}$$



# Application: Distributed Sparse-PCA

- The sparse PCA problem can be formulated as

$$\min_{x} x^T B x + \gamma \|x\|_0, \; \|x\|_2^2 \le 1 \tag{14}$$

  where $-B \succ 0$

- Consider its relaxation

$$\min_{x} x^T B x + \gamma \|x\|_1, \; \|x\|_2^2 \le 1 \tag{15}$$

- Has wide applications, for example large-scale text data analysis

# Application: Distributed Sparse-PCA (cont.)

- In large-scale text analysis, let $C \in \mathbb{R}^{R\times N}$ denote the summary of the text corpora such that
  - A total of $R$ documents (one row one document); $N$ words
  - $C[r, n] = 1$ means word $n$ appears in document $r$

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- Suppose data is divided by $C = [C_1^T, \cdots, C_K^T]^T$, where each $C_k \in \mathbb{R}^{r_k \times N}$ consists of a subset of documents
- $C_k$'s stored on $K$ different agents; no data exchange allowed

$$\min_x -\sum_{k=1}^{K} x^T C_k^T C_k x + \gamma\|x\|_1, \; \|x\|_2^2 \le 1 \tag{17}$$

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$$\min_x -x^T C^T C x + \gamma\|x\|_1,\ \|x\|_2^2 \le 1 \tag{16}$$

- Suppose data is divided by $C = [C_1^T, \cdots, C_K^T]^T$, where each $C_k \in \mathbb{R}^{r_k \times N}$ consists of a subset of documents
- $C_k$'s stored on $K$ different agents; no data exchange allowed

$$\min_x -\sum_{k=1}^{K} x^T C_k^T C_k x + \gamma\|x\|_1,\ \|x\|_2^2 \le 1 \tag{17}$$

- A nonconvex global consensus problem

# The Algorithm

- The augmented Lagrangian function is given by

$$L(\{x_k\}, x; y) = \sum_{k=1}^{K} g_k(x_k) + h(x) + \sum_{k=1}^{K} \langle y_k, x_k - x \rangle + \sum_{k=1}^{K} \frac{\rho_k}{2} \|x_k - x\|^2.$$

**Algorithm 1. The Classical ADMM for the Consensus Problem** (13)

At each iteration $t+1$, compute:

$$x^{t+1} = \underset{x \in X}{\operatorname{argmin}}\, L(\{x_k^t\}, x; y^t) = \operatorname{prox}_{\iota(X)+h}\left[\frac{\sum_{k=1}^{K} \rho_k x_k^t + \sum_{k=1}^{K} y_k^t}{\sum_{k=1}^{K} \rho_k}\right].$$

Each node $k$ computes $x_k$ by solving:

$$x_k^{t+1} = \arg\min_{x_k} g_k(x_k) + \langle y_k^t, x_k - x^{t+1} \rangle + \frac{\rho_k}{2} \|x_k - x^{t+1}\|^2.$$

Each node $k$ updates the dual variable:

$$y_k^{t+1} = y_k^t + \rho_k \left(x_k^{t+1} - x^{t+1}\right).$$

# Outline

# Outline

# Assumptions

**Assumption A.**

A1. $g_k$'s Lipschitz continuous:

$$\|\nabla_k g_k(x_k) - \nabla_k g_k(z_k)\| \leq L_k \|x_k - z_k\|, \ \forall \ x_k, z_k, \ k = 1, \cdots, K.$$

Moreover, $h$ is convex (possible nonsmooth); $X$ is a closed convex set.

A2. $\rho_k$ is large enough such that:

1. For all $k$, the $x_k$ subproblem is strongly convex with modulus $\gamma_k(\rho_k)$;
2. For all $k$, $\rho_k \gamma_k(\rho_k) > 2L_k^2$ and $\rho_k \geq L_k$.

A3. $f(x)$ is bounded from below over $X$.

**Comments:**

- As $\rho_k$ inceases, $x_k$ subproblem eventually becomes strongly convex
- By construction, the $x$ subproblem is also strongly convex
- No assumption on the iterates generated by the algorithm

# Proof Steps

- Use $L(x, \{x_k\}; y)$ as the potential function
- Use a three-step approach
- **Step 1**: Sufficient Descent

$$
\begin{aligned}
& L(x^{t+1}, \{x_k^{t+1}\}; y^{t+1}) - L(x^t, \{x_k^t\}; y^t) \\
& \le -\sigma_0 \|x^{t+1} - x^t\|^2 - \sum_{k=1}^{K} \sigma_k \|x_k^{t+1} - x_k^t\|^2
\end{aligned}
$$

- **Step 2**: Show that $L(x^{t+1}, \{x_k^{t+1}\}; y^{t+1})$ is lower bounded
- **Step 3**: Show that $\{x^{t+1}, \{x_k^{t+1}\}, y^{t+1}\}$ converges to a stationary solution of the global consensus problem

# S1: Sufficient Descent

- To show sufficient descent, we need the following lemma
- **Lemma**: The following is true

$$L_k^2\|x_k^{t+1} - x_k^t\|^2 \geq \|y_k^{t+1} - y_k^t\|^2, \ \forall\ k = 1, \cdots, K, \tag{18}$$

- From the $x_k$ update step, we have the following optimality condition

$$\nabla g_k(x_k^{t+1}) + y_k^t + \rho_k(x_k^{t+1} - x^{t+1}) = 0, \ \forall\ k. \tag{19}$$

which implies

$$\nabla g_k(x_k^{t+1}) = -y_k^{t+1}, \ \forall\ k. \tag{20}$$

- Using the Lipschitz continuity assumption on $\nabla g_k$ we are done

# S1: Sufficient Descent (cont.)

- Using the previous lemma, we can show the following

$$L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^t\}, x^t; y^t)$$
$$\leq \sum_{k=1} K \left( \frac{L_k^2}{\rho_k} - \frac{\gamma_k(\rho_k)}{2} \right) \|x_k^{t+1} - x_k^t\|^2 - \frac{\gamma}{2} \|x^{t+1} - x^t\|^2. \qquad (21)$$

where $\gamma = \sum_{k=1}^K \rho_k$

- First split the successive difference of the augmented Lagrangian by

$$L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^t\}, x^t; y^t)$$
$$= \left( L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^{t+1}\}, x^{t+1}; y^t) \right)$$
$$+ \left( L(\{x_k^{t+1}\}, x^{t+1}; y^t) - L(\{x_k^t\}, x^t; y^t) \right) \qquad (22)$$

# S1: Sufficient Descent (cont.)

- The red term can be bounded by

$$
\begin{aligned}
& L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^{t+1}\}, x^{t+1}; y^t) \\
& = \sum_{k=1}^{K} \langle y_k^{t+1} - y_k^t, x_k^{t+1} - x^{t+1} \rangle = \sum_k \rho_k \|x_k^{t+1} - x^{t+1}\|^2 \\
& = \sum_k \frac{1}{\rho_k} \|y_k^{t+1} - y_k^t\|^2 \le \sum_k \frac{L_k^2}{\rho_k} \|x_k^{t+1} - x_k^t\|^2 
\end{aligned} \tag{23}
$$

- The blue term can be bounded by (using per-block strong convexity)

$$
\begin{aligned}
& L(\{x_k^{t+1}\}, x^{t+1}; y^t) - L(\{x_k^t\}, x^t; y^t) \\
& = L(\{x_k^{t+1}\}, x^{t+1}; y^t) - L(\{x^t{}_k\}, x^{t+1}; y^t) \\
& \quad + L(\{x^t{}_k\}, x^{t+1}; y^t) - L(\{x_k^t\}, x^t; y^t) \\
& \le -\sum_k \frac{\gamma_k(\rho_k)}{2} \|x_k^{t+1} - x_k^t\|^2 - \frac{\gamma}{2} \|x^{t+1} - x^t\|^2,
\end{aligned} \tag{24}
$$

# S2: Lower Bounds

- We then show that for some constant $L^*$

$$\lim_{t\to\infty} L(\{\{x_k^t\}, x^t, y^t\} = L^* > -\infty \tag{25}$$

- We have the following inequalities

$$
\begin{aligned}
& L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) \\
& = h(x^{t+1}) + \sum_{k=1}^K g_k(x_k^{t+1}) + \langle y_k^{t+1}, x_k^{t+1} - x^{t+1}\rangle + \frac{\rho_k}{2}\|x_k^{t+1} - x^{t+1}\|^2 \\
& = h(x^{t+1}) + \sum_{k=1}^K g_k(x_k^{t+1}) + \langle \nabla g_k(x_k^{t+1}), x^{t+1} - x_k^{t+1}\rangle + \frac{\rho_k}{2}\|x_k^{t+1} - x^{t+1}\|^2 \\
& \overset{\text{(a)}}{\geq} h(x^{t+1}) + \sum_{k=1}^K g_k(x^{t+1}) = f(x^{t+1})
\end{aligned}
\tag{26}
$$

where (a) comes from the Lipschitz continuity assumption, and the fact that $\rho_k \geq L_k$ for all $k = 1, \cdots, K$

- By assumption A3, $f(x)$ is lower bounded over $x \in X$

# S3: Convergence

- **Theorem** Assume that Assumption A is satisfied. Then we have the following

    1. We have $\lim_{t\to\infty} \|x_k^{t+1} - x^{t+1}\| = 0,\ k = 1, \cdots, K$
    2. Let $(\{x_k^*\}, x^*, y^*)$ denote any limit point of the sequence $\{\{x_k^{t+1}\}, x^{t+1}, y^{t+1}\}$ generated by Algorithm 1, then it is a stationary solution of problem (13)
    3. If $X$ is a compact set, then the sequence of iterates generated by Algorithm 1 converges to the set of stationary solutions of problem (13)

# S3: Convergence (cont.)

- To show the first item, notice that

$$
\begin{aligned}
& L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^t\}, x^t; y^t) \\
& \le \sum_k \left( \frac{L_k^2}{\rho_k} - \frac{\gamma_k(\rho_k)}{2} \right) \|x_k^{t+1} - x_k^t\|^2 - \frac{\gamma}{2}\|x^{t+1} - x^t\|^2
\end{aligned}
$$

- So the lower boundedness of L implies that

$$
\|x^{t+1} - x^t\| \to 0, \quad \|x_k^{t+1} - x_k^t\| \to 0, \ \forall\ k = 1, \cdots, K. \tag{27}
$$

- By the Lemma, we further obtain $\|y_k^{t+1} - y_k^t\| \to 0$ for all $k = 1, 2, ..., K$, which implies that $\|x_k^{t+1} - x^{t+1}\| \to 0$
- The rest of the proof is checking KKT condition, omitted

# Outline

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# Extensions to Flexible Update Rules

- The above basic analysis can be extended to an implementation with flexible update rules
- Let $k = 1, 2, ..., K$ be the indices for the primal variable blocks $x_1, x_2, ..., x_k$ and $k = 0$ be the index for primal variable block $x$
- Let $\mathcal{C}^t \subseteq \{0, 1, \cdots, K\}$ denote the set of variables updated in iteration $t$, then
  1. **Randomized update rule**: At each iteration $t+1$, a variable block $k$ is chosen randomly with probability $p_k^{t+1}$,

$$\Pr\left(k \in \mathcal{C}^{t+1} \mid x^t, y^t, \{x_k^t\}\right) = p_k^{t+1} \geq p_{\min} > 0. \tag{28}$$

  2. **Essentially cyclic update rule**: There exists a given period $T \geq 1$ during which each index is updated at least once, i.e.

$$\bigcup_{i=1}^{T} \mathcal{C}^{t+i} = \{0, 1, \cdots, K\}, \ \forall\ t. \tag{29}$$

     We call this update rule a period-$T$ essentially cyclic update rule

# Extensions to Flexible Update Rules

**Algorithm 2. The Flexible ADMM for the Consensus Problem** (13)

At each iteration $t+1$, pick an index set $\mathcal{C}^{t+1} \subseteq \{0, \cdots, K\}$.
**If** $0 \in \mathcal{C}^{t+1}$, compute:

$$x^{t+1} = \underset{x \in X}{\arg\min}\, L(\{x_k^t\}, x; y^t) = \operatorname{prox}_{\iota(X)+h}\left[\frac{\sum_{k=1}^K \rho_k x_k^t + \sum_{k=1}^K y_k^t}{\sum_{k=1}^K \rho_k}\right].$$

**Else** $x^{t+1} = x^t$.
**If** $k \in \mathcal{C}^{t+1}$, node $k$ computes $x_k$ by solving:

$$x_k^{t+1} = \arg\min_{x_k} g_k(x_k) + \langle y_k^t, x_k - x^{t+1}\rangle + \frac{\rho_k}{2}\|x_k - x^{t+1}\|^2.$$

Update the dual variable:

$$y_k^{t+1} = y_k^t + \rho_k\left(x_k^{t+1} - x^{t+1}\right).$$

**Else** $x_k^{t+1} = x_k^t$, $y_k^{t+1} = y_k^t$.

# Extension to Proximal Update Rules

- We can also include proximal update for the $x_k$ subproblems
- **Motivation**: in some applications cheap iterations are preferred

**Algorithm 3. A Flexible Proximal ADMM for** (13)

At each iteration $t+1$, compute: $x^{t+1} = \mathrm{argmin}_{x\in X} \; L(\{x_k^t\}, x; y^t)$.
Pick a set $\mathcal{C}^{t+1} \subseteq \{1, \cdots, K\}$.
**If** $k \in \mathcal{C}^{t+1}$, update $x_k$ by solving:

$$x_k^{t+1} = \operatorname*{argmin}_{x_k} \; \langle \nabla g_k(x^{t+1}), x_k - x^{t+1} \rangle + \langle y_k^t, x_k - x^{t+1} \rangle + \frac{\rho_k + L_k}{2} \|x_k - x^{t+1}\|^2.$$

Update the dual variable: $y_k^{t+1} = y_k^t + \rho_k \left( x_k^{t+1} - x^{t+1} \right)$
**Else** let $x_k^{t+1} = x_k^t$, $y_k^{t+1} = y_k^t$.

# Extension to Proximal Update Rules (cont.)

- **Assumption B.** The stepsize $\rho_k$ is chosen large enough such that:

$$\beta_k := \rho_k - 3L_k - \left(\frac{4L_k}{\rho_k^2} + \frac{1}{\rho_k}\right) 2L_k^2 > 0 \quad (30)$$

$$\alpha_k := \frac{\rho_k}{2} - T^2 \left(\frac{4L_k}{\rho_k^2} + \frac{1}{\rho_k}\right) 8L_k^2 > 0 \quad (31)$$

$$\rho_k \geq 5L_k, \; k = 1, \cdots, K. \quad (32)$$

- **Main Result**: Suppose that Assumptions A1, A3 and B hold. Then the following is true for Algorithm 3.
  1. We have $\lim_{t\to\infty} \|x^{t+1} - x_k^{t+1}\| = 0$, $k = 1, \cdots, K$.
  2. Any limit point generated by Algorithm 3 is a stationary solution of problem (13).
  3. If $X$ is a compact set, then the sequence converges to the set of stationary solutions

# Extension to the Sharing Problem

- Consider the following well-known sharing problem

$$\begin{aligned} \min \quad & f(x_1, \cdots, x_K) := \sum_{k=1}^{K} g_k(x_k) + \ell\left(\sum_{k=1}^{K} A_k x_k\right) \\ \text{s.t.} \quad & x_k \in X_k,\ k = 1, \cdots, K, \end{aligned} \tag{33}$$

- $x_k \in \mathbb{R}^{N_k}$ is the variable associated with a given agent $k$
- The variables are coupled through the function $\ell(\cdot)$
- Consider the following reformulation

$$\begin{aligned} \min \quad & \sum_{k=1}^{K} g_k(x_k) + \ell\left(x\right) \\ \text{s.t.} \quad & \sum_{k=1}^{K} A_k x_k = x, \quad x_k \in X_k,\ k = 1, \cdots, K. \end{aligned} \tag{34}$$

- **Multi-block reformulation**, classical ADMM convergent for convex cases?

# Extension to the Sharing Problem (cont.)

**Assumption C.**

C1. $\ell(\cdot)$ Lipschitz continuous with constant $L$. Moreover, $X_k$'s are closed convex sets; each $A_k$ is full column rank, with $\rho_{\min}(A_k^T A_k) > 0$.

C2. The stepsize $\rho$ is chosen large enough such that:

(1) Each $x_k$ and $x$ subproblem is strongly convex, with modulus $\{\gamma_k(\rho)\}_{k=1}^K$ and $\gamma(\rho)$, respectively.

(2) $\rho\gamma(\rho) > 2L^2$, and that $\rho \geq L$.

C3. $f(x_1, \cdots, x_K)$ is lower bounded over $\prod_{k=1}^K X_k$.

C4. $g_k$ is either smooth nonconvex or convex (possibly nonsmooth). For the former case, $g_k$ is Lipschitz continuous over $X_k$ with constant $L_k$.

- **Convergence results similar as before.**
- If the objective is convex, then the stepsize becomes $\rho \geq \sqrt{2}L$

# Outline

# Outline

# Numerical Results

- We consider a special case of the consensus problem

$$\begin{aligned} \min \quad & \sum_{k=1}^{K} x^T B_k x + \lambda \|x\|_1 \\ \text{s.t.} \quad & \|x\|_2^2 \leq 1, \end{aligned} \tag{35}$$

- Each step closed-form
- $N = 1000$, $K = 10$, $\lambda = 100$. Each $B_k = -\xi\xi^H$, where $\xi \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- $\rho_k$ is chosen according to the rule specified in Assumption A2
- We run both the classical and the randomized versions of ADMM, and for the latter case we choose $p_k^t = 0.9$ for all $k, t$
- We also run the classical ADMM with small stepsizes $\hat{\rho}_k = \rho_k / 1000,\ \forall\ k$

# Numerical Results

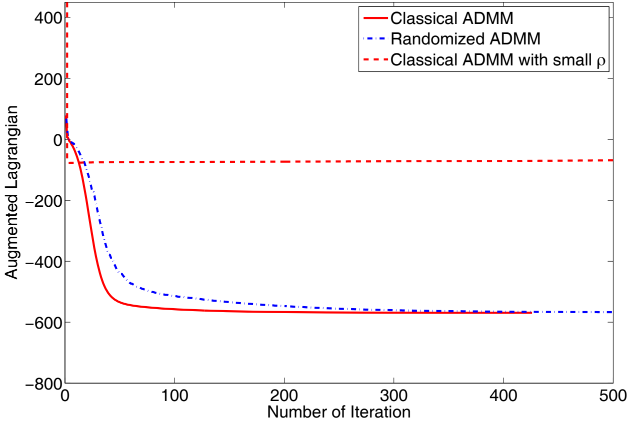



Figure: The value of $L(x^t; y^t)$ for different algorithms.

# Numerical Results

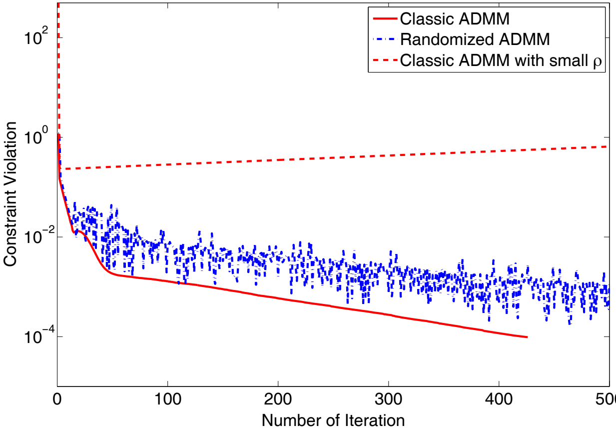



Figure: The value of $\max_k \{\|x_k^t - x_0^t\|\}$ for different algorithms.

# Outline

# Outline

# Weakening the Assumptions

- Key Questions to Address: How to relax the three main assumptions
  1. The Lipschitz continuity assumption on the gradient of the objective function for the last block to be updated

- **Example** $\min \|Z - X^T Y\|^2$, s.t. $X = Y$

- Without additional assumptions, the objective of the last subproblem is NOT globally Lipschitz continuous

- **Example** $\min f(x) + \|y\|_1$, s.t. $x = y$, $f$ is nonsmooth as well (e.g., has constraints)

# Weakening the Assumptions

- Key Questions to Address: How to relax the three main assumptions
  1. The Lipschitz continuity assumption on the gradient of the objective function for the last block to be updated
  2. Taking into consideration nonconvex constraints

- **Example** Consider $\min \|Y - X_1X_2\|^2,\ X_1 \geq 0, X_2 \geq 0$

- Equivalent to

$$\min \|Y - Z\|^2,\ Z = X_1X_2,\ X_1 = \hat{X}_1,\ X_2 = \hat{X}_2,\ \hat{X}_1 \geq 0,\ \hat{X}_2 \geq 0$$

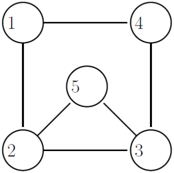
# Weakening the Assumptions

- Key Questions to Address: How to relax the three main assumptions
  1. The Lipschitz continuity assumption on the gradient of the objective function for the last block to be updated
  2. Taking into consideration nonconvex constraints
  3. General inequality constraints

- **Example** The consensus problem on a graph (no central controller)

$$\min_{k} \sum_{k=1}^{K} g_k(x_k) \text{ s.t. } Ax = 0 \tag{36}$$

where the matrix $A$ is the edge-node incidence matrix




$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix}.$$

# Thank You!